Presentation Flow (15 Minutes)

👤 Speaker 1 (Introduction) – 2.5 minutes

Slides 2–4

Opening

"Good morning everyone. We are Group 1, and today we are presenting on **Personal Location Data: Ethics and Governance**, based on the Royal Society and British Academy report on Data Governance."

"Before we begin, let me ask you a simple question."

Ask the audience

"How many of you use Google Maps, Uber, Strava, Apple Maps or fitness trackers?"

(Pause for 2 seconds.)

"Every time we use these apps, they collect our location data. This data makes our lives easier, but it also raises important questions about privacy and ethics. That's what today's presentation is about."

Explain the agenda

"We'll first explain what personal location data is, then discuss its benefits, challenges, governance, real-life examples, solutions, and finally conclude with key takeaways."

Explain the paper

"This report was written by Michael Veale in 2017 before GDPR came into force. The main question the paper asks is:
How can society enjoy the benefits of location data while protecting people's privacy?"

Self-tracking

Briefly explain:

- Private
- Pushed
- Imposed
- Communal
- Exploited

Finish with

"Now my teammate will explain how location data benefits society."

👤 Speaker 2 (Benefits) – 2 minutes

Speaker 2 Script (Slides 5–6)

Slide 5– Opportunities: Using Location Data for Good

"Thank you."

"Now I'll explain how location data can be used in positive ways to improve our daily lives and support better decision-making."

"One of the biggest advantages of location data is **evidence-based urban planning**. Instead of relying only on surveys or manual traffic counts, governments can use real-time location data to understand how people travel."

"For example, **Strava Metro** collects cycling data from users. This information helps local governments identify the most popular cycling routes and decide where new cycle lanes or cycling infrastructure should be built. In London alone, over seven million cycling journeys were uploaded in just one year."

"Another example is **Uber Movement**. Uber shares anonymous journey information with city planners, allowing them to analyse traffic patterns, understand congestion, and improve road planning without revealing individual users' identities."

"Similarly, **Waze** uses location data from drivers to provide live traffic updates, accident reports and the fastest routes. It also works with many cities to improve traffic management and reduce congestion."

"Compared with traditional surveys, these digital datasets are much cheaper, cover a much larger number of people, and provide information in real time. This helps governments make faster and more informed decisions."

"Location data also supports the development of **smart cities**, where technology is used to improve transport systems, public services and even healthcare. For example, monitoring elderly people through location-based health applications can help them live independently while ensuring their safety."

Slide 6 – What Can Be Inferred from Location Data?

"Although location data provides many benefits, it can also reveal much more about us than most people realise."

"Even if companies only collect our location, they can use that information to make many inferences."

"For example, they can identify where we **live**, where we **work**, our daily routines, the places we visit most often, and even estimate our **income** based on the neighbourhoods we spend time in."

"Location patterns can also reveal our **behaviour**, such as where we shop, how we travel, our hobbies, and the places we regularly visit. With advanced machine learning, researchers have shown that it may even be possible to predict certain aspects of a person's personality and future movements."

"One of the most surprising findings mentioned in the paper is that researchers discovered **just four location points are enough to uniquely identify about 95% of people**. This means that even data that appears anonymous can often be linked back to a specific individual."

"This shows that while location data offers many valuable opportunities, it also creates significant privacy risks."

Transition

"Now that we've seen both the benefits and the risks of location data, my teammate will explain why **consent** is one of the biggest ethical challenges in collecting and using this information."

👤 Speaker 3 (Consent & Privacy) – 2.5 minutes

Slides 7–8

Start

"Many people believe they choose to share their location, but in reality consent is often not genuine."

Explain simply:

- GPS isn't the only tracker.
- Wi-Fi and Bluetooth also track.
- Insurance forces people.
- Friends can reveal your location.
- Elderly people may not fully understand consent.

Then move to bias.

Explain

Strava

- Mostly male users
- Richer areas
- Sports cyclists

Waze

- Criminals tracked police
- Fake traffic reports
- Traffic diverted into residential areas

End

"These examples show that location data can sometimes create unfair or even dangerous outcomes."

👤 Speaker 4 (Governance & Solutions) – 2.5 minutes

Slides 9–10

Start

"So how is location data regulated?"

Explain GDPR

- Personal data
- Consent
- Right to delete
- Right to access

Mention loopholes.

Then explain solutions

Very briefly:

- Better privacy protection
- Clear consent
- Alerts when location is shared
- Reduce bias
- Safety checks
- Better regulations

Finish

"These solutions try to balance innovation with privacy."

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👤 Speaker 5 (Real-World Examples & Ethical Framework) – 2.5 minutes

Slides 11–14

Start

"Now let's look at some real-world examples that show what can happen when location data is not managed properly."

Slide 11 – Case Study Analysis

"This slide highlights what companies have done well and where improvements are needed."

"Some good practices include Strava providing demographic information to help users understand bias, Uber sharing traffic data with city planners, Waze working with cities to improve roads, and GDPR recognising location data as personal data."

"However, there are also weaknesses. Many companies do not clearly explain what they can infer from users' location data. Consent is often just a tick-box rather than truly informed, and demographic information is usually optional instead of mandatory."

"This shows that stronger regulations and better transparency are still needed."

Slide 12 – Real-World Examples

"Here are four examples of location data being misused."

- **Waze:** Criminals were able to track police locations, raising public safety concerns.
- **OkCupid:** A researcher published data from 70,000 public dating profiles without users' permission, causing a major privacy controversy.
- **Strava Heatmap:** Fitness tracking data accidentally revealed the locations and patrol routes of secret military bases.
- **Apple & Google:** Both companies faced criticism after reports showed that smartphones were storing users' location history without many people realising it.

"These cases show that even useful technology can create serious privacy and security risks."

Slide 13 – Ethical Framework

"The biggest ethical challenge is finding the right balance."

"On one side, location data creates public benefits such as better transport planning, smarter cities and improved emergency services."

"On the other side, it raises concerns about privacy, surveillance, power imbalance and bias."

"The challenge is to gain the benefits without compromising people's rights."

Slide 14 – Solutions

"To address these challenges, both technical and regulatory solutions are needed."

"Technical solutions include differential privacy, permission dashboards and better data protection tools."

"Policy solutions include stronger consent requirements, safety audits, demographic reporting and better regulations for companies using location data."

Transition

"Now I'll hand over to my teammate, who will explain how different organisations work together to govern location data and conclude our presentation."

👤 Speaker 6 (Comparative Analysis, Conclusion & Q&A) – 2.5–3 minutes

Slides 15–17

Slide 15 – Comparative Analysis

"Thank you. I'll finish our presentation by looking at how different groups help govern location data."

Industry

"Companies such as Strava, Uber and Waze have introduced their own privacy measures. However, these are voluntary and may not be applied consistently."

Regulators

"Governments and regulators, through laws like GDPR, protect users by treating location data as personal data and giving people rights over their information."

Researchers

"Researchers identify new privacy risks, such as how just a few location points can identify a person, and they help improve future policies."

Technology Community

"Technology experts develop privacy-enhancing tools like differential privacy and secure data-sharing methods, but these are not yet mandatory for all companies."

"This comparison shows that protecting location data is a shared responsibility. No single group can solve these challenges alone."

Slide 16 – Conclusion

"To conclude, our presentation highlights four main points."

"First, location data provides many benefits, including better transport, city planning and smarter public services."

"Second, it also creates serious privacy, security and ethical concerns because it can reveal sensitive information about people's lives."

"Third, consent is often not fully informed, and location datasets can be biased, leading to unfair decisions."

"Finally, stronger governance, greater transparency and responsible data management are essential to protect individuals while still allowing society to benefit from location data."

Final Message

"The biggest message from this paper is that the greatest risk is not only hackers—it is the continuous collection and analysis of our everyday location data without us fully understanding how it is being used."

Slide 17 – Thank You

"Thank you everyone for your time and attention."

"We hope this presentation has helped you understand both the opportunities and the ethical challenges of personal location data."

"We are now happy to answer any questions."

Possible Questions from Lecturer

Q1. What is personal location data?

Answer

Information that shows where a person is or has been, collected using GPS, smartphones, apps and wearable devices.

Q2. Why is location data useful?

Answer

It improves traffic, city planning, emergency services and navigation.

Q3. What is the biggest privacy risk?

Answer

Re-identification. Even anonymous location data can identify individuals.

Q4. What is GDPR?

Answer

A European law that protects personal data, including location data.

Q5. Why is Strava biased?

Answer

Most users are male sports cyclists, so the data does not represent everyone.

Speaker	Slide No.	Slide Title	Topic Covered	Time
1	**1–3**	Title, Agenda, What is This Paper About?	Introduction, overview of the paper, importance of location data, five modes of self-tracking	2.5 min
2	**4–5**	Opportunities & What Can Be Inferred from Location Data	Benefits of location data, smart cities, Strava, Uber, Waze, privacy risks from location data	2–2.5 min
3	**6–7**	Consent Issues & Bias in Location Data	Consent challenges, hidden tracking, insurance tracking, Strava bias, Waze issues	2.5 min
4	**8–10**	Governance & Solutions	GDPR, governance, privacy protection, technical and regulatory solutions	2.5 min
5	**11–14**	Case Study Analysis, Real-World Examples, Ethical Framework, Technical & Regulatory Solutions	What companies did right/wrong, Waze, OkCupid, Strava, Apple, ethical issues, balancing public benefit and privacy	2.5–3 min
6	**15–17**	Comparative Analysis, Conclusion & Q&A	Role of industry, regulators, researchers and technology community, final conclusions, thank you, questions	2.5–3 min